MATLAB for Computer Networking

Programming Exercises and Network Analysis

12 Exercises	Networking Concepts	MATLAB Practice
Clearly defined tasks and supplied data	Addressing, routing, reliability, and performance	Calculations, tables, and plots

Student Instructions

- Write a MATLAB script or function for each exercise.
- Use the given values and formulas. Do not hard-code calculated answers.
- Display results clearly, with suitable labels and units.
- For each requested plot, include a title, labelled axes, and a grid; add a legend when comparing multiple series.
- Briefly explain what the calculated results or graph show.
- Submit your MATLAB code and the required tables/plots as instructed by your course instructor.

Prepared for Computer Networking coursework

Question 1 — Network Throughput and Data Transfer Time

Problem scenario

A department transfers a 24 MB file from one computer to another over a network link. The link has a nominal capacity of 20 Mbps, but protocol overhead and other traffic mean that only 75% of the link capacity is available for file data.

Given data and assumptions

• File size = 24 MB, where 1 MB = 1,000,000 bytes.

• Link capacity = 20 Mbps.

• Usable capacity factor = 0.75.

• Ignore connection setup time and retransmissions.

Tasks to complete

1. Create MATLAB variables for the file size, link capacity, and usable capacity factor.
2. Calculate the effective data rate in Mbps.
3. Convert the file size to bits and calculate the estimated transfer time in seconds.
4. Repeat the calculation for usable capacity factors of 0.50, 0.60, 0.75, 0.90, and 1.00.
5. Plot usable capacity factor against estimated transfer time. Add a title, axis labels, grid, and appropriate units.
6. Display the effective data rate and transfer time clearly in the Command Window.

Expected submission: A MATLAB script, a small results table, and one labelled plot.

Question 2 — Transmission Delay, Propagation Delay, and Total Delay

Problem scenario

A 1,500-byte packet is sent over a point-to-point link. The packet experiences transmission delay, propagation delay, and processing delay. For this exercise, assume there is no queueing delay.

Given data and assumptions

• Packet size = 1,500 bytes.

• Link bandwidth = 10 Mbps.

• Link distance = 1,200 km.

• Signal propagation speed = 2 × 10^8 metres/second.

• Processing delay at the sender and receiver combined = 2 ms.

• Use transmission delay = packet size in bits / bandwidth; propagation delay = distance / propagation speed.

Tasks to complete

1. Calculate packet size in bits.
2. Calculate transmission delay in seconds and milliseconds.
3. Convert the link distance to metres and calculate propagation delay.
4. Calculate total one-way delay by adding transmission, propagation, and processing delays.
5. Repeat the calculation for bandwidths of 5, 10, 20, and 50 Mbps while keeping all other values constant.
6. Plot bandwidth against total one-way delay and briefly explain why the delay does not become zero as bandwidth increases.

Expected submission: A MATLAB script with labelled calculations, a comparison table, and one plot.

Question 3 — Stop-and-Wait ARQ Efficiency

Problem scenario

A sender uses Stop-and-Wait ARQ. It sends one frame and waits for an acknowledgement before sending the next frame. Assume the acknowledgement transmission time is negligible and no frames are lost in the first calculation.

Given data and assumptions

• Frame size = 1,000 bytes.

• Link bandwidth = 4 Mbps.

• One-way propagation delay = 20 ms.

• Ignore processing delay and acknowledgement transmission time.

• Use efficiency = frame transmission time / (frame transmission time + round-trip propagation time).

Tasks to complete

1. Calculate the frame transmission time in seconds and milliseconds.
2. Calculate the round-trip propagation time.
3. Calculate Stop-and-Wait efficiency as a percentage.
4. Estimate the effective throughput in Mbps by multiplying efficiency by the link bandwidth.
5. Repeat the calculation for one-way propagation delays of 5, 10, 20, 40, and 80 ms.
6. Plot propagation delay against efficiency and explain the trend in two or three sentences.

Expected submission: A MATLAB script, printed numerical results, and one labelled plot.

Question 4 — CRC Error Detection

Problem scenario

A sender uses a Cyclic Redundancy Check (CRC) to detect bit errors. Write a MATLAB script that performs modulo-2 binary division using XOR operations.

Given data and assumptions

• Dataword = 1101011011.

• Generator bit pattern = 10011.

• Append (number of generator bits – 1) zeros to the dataword before division.

• The CRC remainder length equals the number of appended zeros.

Tasks to complete

1. Store the dataword and generator as row vectors containing only 0s and 1s.
2. Implement modulo-2 long division. When the current leading bit is 1, XOR the aligned generator bits; when it is 0, move to the next bit.
3. Find and display the CRC remainder.
4. Append the remainder to the original dataword to form the transmitted codeword.
5. Divide the codeword by the generator at the receiver and verify that the remainder is all zeros.
6. Flip one bit of the codeword (the 5th bit from the left) and repeat the receiver check. State whether the error is detected.

Expected submission: A MATLAB script that displays the remainder, codeword, and receiver verification results.

Question 5 — IPv4 Subnet Calculation

Problem scenario

A college laboratory receives the IPv4 block 192.168.10.0/24. The block must be divided into four equal-sized subnets for four rooms. Each room needs at least 50 usable host addresses.

Given data and assumptions

• IPv4 address length = 32 bits.

• Number of addresses in a subnet = 2^(32 – prefix length).

• For this exercise, usable host addresses = total addresses – 2.

• Network and broadcast addresses cannot be assigned to ordinary hosts.

Tasks to complete

1. Calculate the minimum number of host bits required to support at least 50 usable hosts in one subnet.
2. Determine the prefix length and total address count for each subnet.
3. Check whether the /24 block can provide four equal-sized subnets of the required size.
4. List the network address, first usable address, last usable address, and broadcast address for all four subnets.
5. Write a MATLAB function that accepts an IPv4 address and prefix length and returns the network and broadcast addresses. Test it using 192.168.10.77/26.
6. Display the subnet information in a clearly labelled table.

Expected submission: A MATLAB script or function and a subnet-allocation table.

Question 6 — Routing with a Link-Cost Matrix

Problem scenario

A network contains five routers: A, B, C, D, and E. Each link is bidirectional. The link cost represents a relative routing cost, and a value of 0 indicates that there is no direct link between two different routers.

Given data and assumptions

• Router order: A, B, C, D, E.

• Cost matrix:

```
[0 2 5 0 0;
2 0 1 3 0;
5 1 0 2 6;
0 3 2 0 1;
0 0 6 1 0]
```

• Treat the matrix as symmetric. The cost from a router to itself is zero.

Tasks to complete

1. Store the cost matrix in MATLAB and display it.
2. Find the lowest-cost route from Router A to each of the other routers. You may implement Dijkstra's algorithm or use a suitable MATLAB graph function.
3. Display the total route cost and the sequence of routers for each destination.
4. Draw the network and label each link with its cost.
5. Highlight the lowest-cost route from A to E on the diagram.
6. Change the cost of link B–D from 3 to 8 and repeat the calculations. Identify any route or cost that changes.

Expected submission: A MATLAB script, routing table, and labelled network diagram.

Question 7 — Pure ALOHA and Slotted ALOHA

Problem scenario

Multiple stations share a communication channel. Compare the theoretical throughput of Pure ALOHA and Slotted ALOHA for different offered loads.

Given data and assumptions

• Let G represent the average number of transmission attempts per frame time.

• Pure ALOHA normalized throughput: $S = G \times \exp(-2G)$.

• Slotted ALOHA normalized throughput: $S = G \times \exp(-G)$.

• Use G values from 0 to 3 in increments of 0.05.

Tasks to complete

1. Create a vector of G values in MATLAB.
2. Calculate normalized throughput for both protocols using the given formulas.
3. Plot both curves on the same axes with a legend, title, axis labels, and grid.
4. Find and display the maximum throughput and the corresponding G value for each protocol.
5. Display the throughput of both protocols at G = 0.5, 1.0, and 2.0.
6. Write a short comparison explaining which protocol achieves the higher theoretical maximum and why.

Expected submission: A MATLAB script, one comparison plot, and a brief written interpretation.

Question 8 — TCP Congestion Window

Problem scenario

A simplified TCP sender starts with a small congestion window. During slow start, the congestion window doubles each round until it reaches a threshold. During congestion avoidance, it increases by one segment per round.

Given data and assumptions

• Initial congestion window (cwnd) = 1 segment.

• Slow-start threshold (ssthresh) = 8 segments.

• Number of rounds = 15.

• At the end of round 10, simulate a loss event: set ssthresh to half the current cwnd (minimum 2 segments) and set cwnd to 1 segment for round 11.

• For this simplified exercise, apply the loss rule exactly as stated and ignore retransmission timers and delayed acknowledgements.

Tasks to complete

1. Create a vector to store cwnd for each of the 15 rounds.
2. Implement slow start by doubling cwnd each round while cwnd is below ssthresh.
3. When cwnd reaches the threshold, increase it by one segment per round.
4. Apply the specified loss event after round 10 and continue the simulation using the updated threshold.
5. Display the cwnd value for each round in a table.
6. Plot round number against cwnd and mark round 10 as the loss event. Label the slow-start and congestion-avoidance phases.

Expected submission: A MATLAB script, a congestion-window table, and one labelled plot.

Question 9 — Packet Loss and Network Performance

Problem scenario

A test program sends 10 packets through a network path. The packet delivery status and measured delay are provided below. Use MATLAB to summarize the network's performance.

Given data and assumptions

• Packet numbers = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

• Delivery status (1 = received, 0 = lost): [1 1 1 0 1 1 0 1 1 1].

• Measured one-way delays in milliseconds for received packets: [12 15 11 — 18 14 — 20 13 16].

• Treat the dash for a lost packet as missing data, not as a delay of zero.

Tasks to complete

1. Store packet numbers, delivery status, and delay data in MATLAB. Use NaN for delays of lost packets.
2. Calculate the number of transmitted, received, and lost packets.
3. Calculate packet delivery ratio and packet loss ratio as percentages.
4. Calculate the mean, median, minimum, and maximum delay using received packets only.
5. Plot packet number against measured delay. Make lost packets visually distinct or indicate them separately.
6. Display all performance metrics with clear labels and briefly explain why lost packets must not be included as zero-delay packets.

Expected submission: A MATLAB script, a performance summary, and one labelled plot.

Question 10 — Bandwidth Comparison and File Download Time

Problem scenario

A user downloads the same file over three network connections. Each connection has a different bandwidth and a different utilization level. Compare the expected download time under a simplified model.

Given data and assumptions

• File size = 150 MB, where 1 MB = 1,000,000 bytes.

• Connection A: bandwidth = 25 Mbps, usable fraction = 0.80.

• Connection B: bandwidth = 50 Mbps, usable fraction = 0.55.

• Connection C: bandwidth = 100 Mbps, usable fraction = 0.30.

• Effective bandwidth = nominal bandwidth × usable fraction. Ignore setup time and retransmissions.

Tasks to complete

1. Create MATLAB arrays for the three connection names, nominal bandwidths, and usable fractions.
2. Calculate effective bandwidth for each connection.
3. Calculate the file size in bits and estimated download time in seconds for each connection.
4. Display a table comparing nominal bandwidth, effective bandwidth, and download time.
5. Plot the estimated download time for each connection using a bar chart.
6. Identify the fastest connection under the given assumptions and explain why the highest nominal bandwidth does not always produce the shortest download time.

Expected submission: A MATLAB script, comparison table, and labelled bar chart.

Question 11 — Router Queue and Waiting Time

Problem scenario

Packets arrive at a router and wait in a first-come, first-served queue for transmission. Use the supplied arrival and service times to calculate each packet's waiting time and total time in the system. This is a deterministic queue exercise, so no random-number generation is required.

Given data and assumptions

• Interarrival times in milliseconds for packets 1–8: [0 3 2 5 1 4 2 3]. The first packet arrives at time 0.

• Service times in milliseconds for packets 1–8: [4 2 3 2 4 1 3 2].

• Arrival time for packet 1 = 0. For later packets, arrival time is the cumulative sum of the interarrival times.

• Service begins when both the packet has arrived and the server is free.

Tasks to complete

1. Calculate the arrival time of every packet.
2. Calculate service-start time using max(arrival time, previous packet's departure time).
3. Calculate departure time by adding service time to service-start time.
4. Calculate waiting time as service-start time minus arrival time.
5. Calculate time in the system as departure time minus arrival time.
6. Display a table containing packet number, arrival time, service start, departure time, waiting time, and time in the system.
7. Plot packet number against waiting time and identify which packet experienced the longest wait.

Expected submission: A MATLAB script, a queueing-results table, and one labelled plot.

Question 12 — Multi-Hop Delay and Bottleneck Link

Problem scenario

A packet travels through three store-and-forward links. Each router forwards the packet only after receiving the entire packet. Calculate the delay for the first packet and estimate the steady-state payload throughput for a continuous stream.

Given data and assumptions

• Link capacities = [10 8 4] Mbps.

• Propagation delays for the three links = [3 5 4] ms.

• Packet payload = 1,000 bytes.

• Header size = 40 bytes per packet.

• Transmission delay on each link = total packet bits / link capacity.

• Ignore queueing, processing delay, acknowledgements, and retransmissions. For a long stream, use bottleneck capacity × payload / (payload + header) as the approximate payload throughput.

Tasks to complete

1. Calculate total packet size, including header, in bytes and bits.
2. Calculate transmission time on each link in seconds and milliseconds.
3. Add all transmission and propagation delays to estimate the one-way delay of the first packet.
4. Identify the bottleneck link and calculate approximate steady-state payload throughput in Mbps.
5. Repeat the payload-throughput calculation for payload sizes of 500, 1,000, 1,500, and 3,000 bytes while keeping the header size constant.
6. Plot payload size against payload efficiency and steady-state payload throughput. Explain briefly why first-packet delay and long-stream throughput are calculated differently.

Expected submission: A MATLAB script, a summary table, and labelled plots.